

MCV4UY - Unit 3 Test - Derivatives

Name _____

Date _____

Total
/42

KNOWLEDGE

1. Differentiate the following using different rules. Simply/factor where possible. (NO negative exponents in final answers).

a. $f(x) = \frac{x^{-7} - x^{-5}}{x^{-3}}$

b. $f(x) = \frac{x^3 + x^2 + x + 1}{x + 1}$

(2)

(2)

c. $y = \frac{\sec x}{7x^2}$

d. $y = \csc x \cot x$

(2)

(2)

2. Find the equation of the tangent line to the curve $y = 2\sin x \cos x$ at $x = \frac{\pi}{2}$.

(3)

3. Calculate the $\frac{d^4y}{dx^4}$ of $f(x) = \frac{2}{3x^6} + \frac{x^4}{5} + \frac{2}{x^3} + \frac{1}{2}x$

(3)

COMMUNICATION

6. Sketch a function that has exactly one point of discontinuity, whose derivative is always positive, but has **two** points where it is **not** differentiable. **Explain your reasoning.**

(3)

7. The figure graphed below shows the velocity of a particle moving along a coordinate line.

a. When is the particle moving right?

(0.5)

b. When does the particle change directions?

(0.5)

c. When is the particle slowing down?

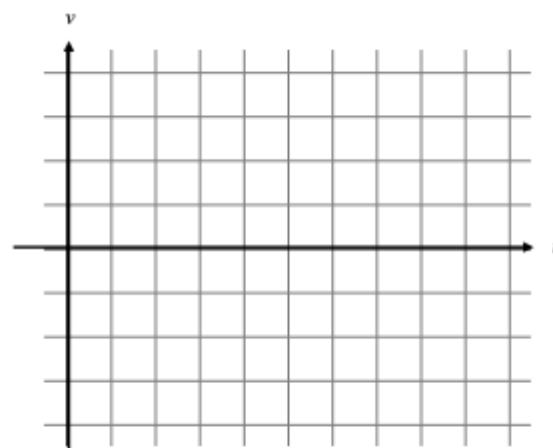
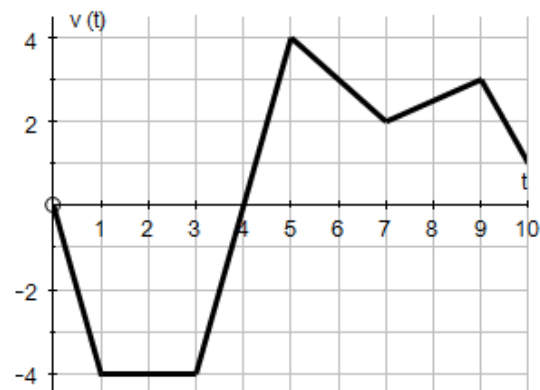
(0.5)

d. When is the particle's acceleration positive?

(0.5)

e. Graph the particle's acceleration for $0 < t < 10$.

(2)



APPLICATION

4. The equation of motion of a particle is $s = t^4 - 2t^3 + t^2 - t$, where s is in meters and t is in seconds.

a. What is the position at 2 seconds?

(1)

b. When does the particle stop moving?

(2)

c. What is the particle's acceleration at 3 seconds?

(2)

d. Describe the motion of the particle from $0 \leq t \leq 2$. At what values of t , does the particle change directions? **Provide a detailed explanation.**

(3)

5. The daily cost, in dollars, if producing x yards of a certain fabric is $C(x) = 1200 + 12x - 0.1x^2 + 0.0005x^3$, and currently they produce 98 yards daily.

a. What is the cost of producing 98 yards?

(1)

b. What would the actual additional cost of increasing production to 99 yards be?

(1)

c. What is the marginal cost of the 201st yard?

(2)

THINKING

8. Find the points on the curve $y = \cot x, 0 < x < 2\pi$, where the tangent line is parallel to the tangent of the curve $y = -x^2 - 3$ at $(1, -4)$.

(4)

9. If $f(x) = (x - a)(x - b)(x - c)$, show that $\frac{f'(x)}{f(x)} = \frac{1}{x-a} + \frac{1}{x-b} + \frac{1}{x-c}$.

(2)

10. The line $x = c$, where $c > 0$, intersects the cubic $f(x) = 2x^3 + 3x^2 - 9$ intersects at P and $g(x) = 4x^2 + 4x + 5$ intersects at Q. If the tangent line to $f(x)$ at P is parallel to the tangent line to $g(x)$ at Q, determine c.

(3)